

Exploring Solvent Effects in the Radiosynthesis of ^{18}F -Labeled Thymidine Analogues toward Clinical Translation for Positron Emission Tomography Imaging

Jindian Li, Juno Van Valkenburgh, Peter S. Conti, and Kai Chen*

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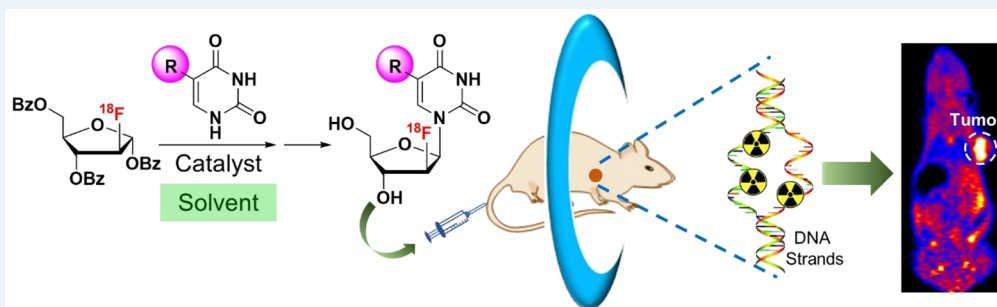
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ABSTRACT: Thymidine analogues, 5-substituted 2'-deoxy-2'- ^{18}F -fluoro-arabinofuranosyluracil derivatives, are promising positron emission tomography (PET) tracers being evaluated for noninvasive imaging of cancer cell proliferation and/or reporter gene expression. We report the radiosynthesis of 2'-deoxy-2'- ^{18}F -fluoro-5-methyl-1- β -D-arabinofuranosyluracil (^{18}F FMAU) and other 2'-deoxy-2'- ^{18}F -fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues using 1,4-dioxane to replace the currently used 1,2-dichloroethane. Compared to 1,2-dichloroethane, 1,4-dioxane is analyzed as a better solvent in terms of radiochemical yield and toxicity concern. The use of a less toxic solvent allows for the translation of the improved approach to clinical production. The new radiolabeling method can be applied to an extensive range of uses for ^{18}F -labeling of other nucleoside analogues.

KEYWORDS: solvent effects, radiosynthesis, ^{18}F -labeled thymidine analogues, clinical translation, PET imaging, cancer

Excessive cellular proliferation is one of many distinct cancer-related hallmarks.¹ A good number of extra-organismal assays have been developed to measure tumor proliferation rates.² However, these assays largely require invasive procedures to remove a small piece of living tissues or a sample of cells from the body, rendering difficulties in assessing tumor proliferation in real time, over the course of treatment, and in multiple regions, particularly for patients with diverse metastatic lesions.³ Molecular imaging has emerged at the forefront in the area of “personalized medicine” to obtain timely and noninvasive evaluation of biological and physiological processes in living bodies and improve our understanding of diseases.^{4,5} Radiofluorinated analogues of 2'-deoxy-2'-fluoro-5-substituted-1- β -D-arabinofuranosyluracil (Figure 1) are promising PET radiotracers for evaluating tumor proliferation and imaging reporter gene expression.^{3,6} The radiotracers are phosphorylated by thymidine kinases TK1 and/or TK2 and further integrated into host DNA (Figure 1).⁷ There is evidence that therapeutic response could be defined earlier and perhaps more accurately by measuring changes in DNA synthesis within tumors.^{8,9} Therefore, 2'-deoxy-2'- ^{18}F -fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues have great potential for use in not only early diagnosis

of diseases, but also identifying treatment effects,^{10,11} and thus assisting in clinical decision-making processes and enabling treatment optimization for individual patients (“personalized medicine”).¹²

^{18}F is one of the most common radionuclides for PET imaging because of its excellent chemical and nuclear-physical properties.¹³ ^{18}F has a half-life of 109.77 min which allows multistep synthesis and longer imaging protocols. In addition, the low β^+ energy of ^{18}F , 0.64 MeV, leads to high resolution PET images due to a short positron linear range in tissue.¹⁴ 2'-Deoxy-2'- ^{18}F -fluoro-5-methyl-1- β -D-arabinofuranosyluracil (^{18}F FMAU) is a promising PET tracer currently being investigated in preclinical studies and clinical trials for evaluating cell proliferation in multiple carcinomas, such as breast carcinoma, prostatic carcinoma, and nonsmall cell lung

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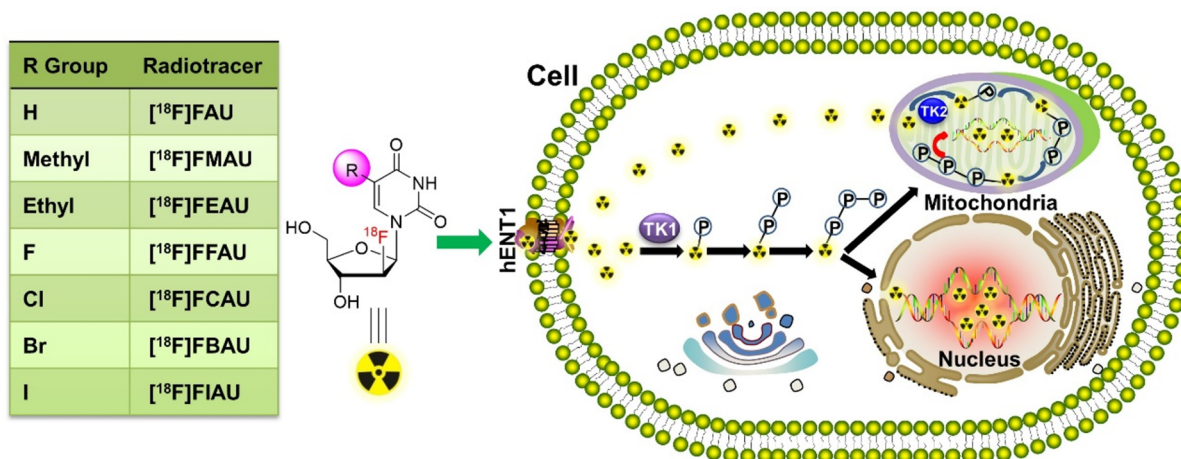
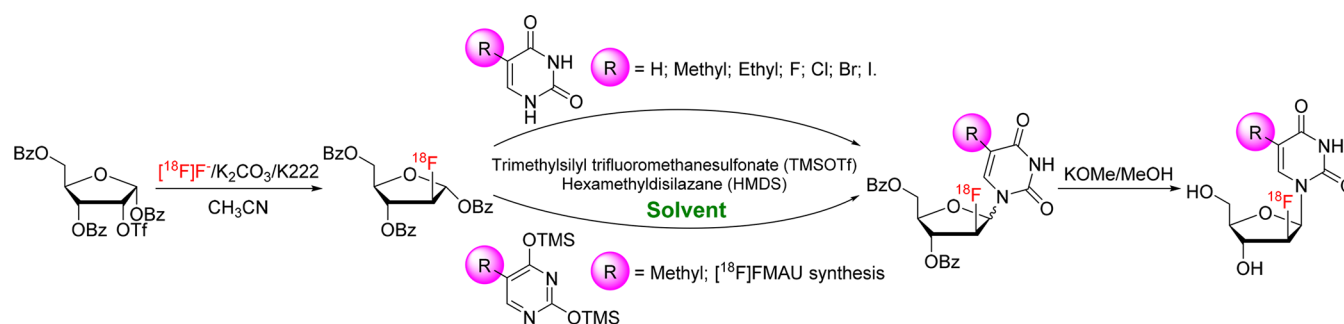


Figure 1. Chemical structures of 2'-deoxy-2'-[¹⁸F]fluoro-5-substituted-1-β-D-arabinofuranosyluracil analogues and their involvement in potential DNA synthesis pathways.

Scheme 1. One-Pot Radiosynthesis of 2'-Deoxy-2'-[¹⁸F]fluoro-5-substituted-1-β-D-arabinofuranosyluracil Analogues



carcinoma.^{15–20} An advantage of [¹⁸F]FMAU over another widely used thymidine analogue, 3'-deoxy-3'-[¹⁸F]-fluorothymidine ([¹⁸F]FLT), is the ability for [¹⁸F]FMAU to incorporate into DNA. Compared with [¹⁸F]FMAU, [¹⁸F]FLT cannot be substantially incorporated into DNA due to the fluorinated 3'-position of deoxyribose acting as a terminator of the growing DNA chain.³ In addition, 2'-deoxy-2'-[¹⁸F]fluoro-5-ethyl-1-β-D-arabinofuranosyluracil ([¹⁸F]FEAU)^{21–23} and 2'-deoxy-2'-[¹⁸F]fluoro-5-iodo-1-β-D-arabinofuranosyluracil ([¹⁸F]FIAU)²⁴ are PET radiotracers for imaging reporter gene herpes virus type 1 thymidine kinase (HSV1-tk) expression. Therefore, they have been used for gene-based therapy, transgenic models, and cell trafficking.²⁵ 2'-Deoxy-2'-[¹⁸F]-fluoro-1-β-D-arabinofuranosyluracil ([¹⁸F]FAU)²⁵ can be phosphorylated and methylated by thymidine kinase and thymidylate synthase, respectively, and then incorporated into DNA.²⁶ Consequently, [¹⁸F]FAU is a promising PET probe for evaluating tumor growth and studying the pharmacokinetics and metabolism of FAU acting as a chemotherapeutic agent.^{26–28} In addition, 2'-deoxy-2'-[¹⁸F]fluoro-5-fluoro-1-β-D-arabinofuranosyluracil ([¹⁸F]FFAU)²⁹ and 2'-deoxy-2'-[¹⁸F]fluoro-5-chloro-1-β-D-arabinofuranosyluracil ([¹⁸F]-FCAU)³⁰ are also promising PET probes for imaging the expression of HSV1-tk genes.

We and other groups reported the radiolabeling of [¹⁸F]FMAU and its thymidine analogues,^{31,32} involving the radiosynthesis of 2-[¹⁸F]fluoro-1,3,5-tri-O-benzoyl arabinofuranose and its conversion to 1-bromo-2-[¹⁸F]fluoro-1,3,5-tri-O-benzoyl arabinofuranose. The latter could be coupled to various 2,4-bis-trimethylsilyluracil derivatives. Hydrolysis of the

protecting groups from the sugar moiety provided the desired products. However, this method of making 2'-deoxy-2'-fluoro-5-substituted-1-β-D-arabinofuranosyluracil analogues is rather tedious involving multistep procedures leading to a low radiochemical yield of desired products and inconvenience for clinical use. The synthetic approach using Friedel–Crafts catalysts was previously reported by our group to simplify synthesis conditions and shorten the reaction time.^{33–35} However, a very toxic solvent, 1,2-dichloroethane (DCE), was employed as the solvent in the coupling of 2-deoxy-2-[¹⁸F]fluoro-1,3,5-tri-O-benzoyl-D-arabinofuranose (¹⁸F-labeled sugar) and uracil bases. In the United States Pharmacopeia (USP) General Chapter <467>, DCE is defined as a Class 1 residual solvent, and its injectable concentration is limited at 5 ppm (ppm) due to its highly toxic potential to humans. The residual DCE in the PET drug injection is strictly controlled by the US Food and Drug Administration (FDA). In addition, the quantitation limit of an extremely low concentration of residual solvents, such as DCE (≤5 ppm), puts forward a huge challenge for the method validation of gas chromatography. Therefore, finding a suitable solvent for the radiosynthesis of [¹⁸F]FMAU and its analogues is in urgent demand for paving the way of their clinical translation.

In the current research, the feasibility of using other polar and nonpolar solvents was analyzed in the coupling of ¹⁸F-labeled sugar and 5-substituted uracil using trimethylsilyl trifluoromethanesulfonate (TMSOTf) and hexamethyldisilazane (HMDS) (Scheme 1). After the identification of 1,4-dioxane as the optimal solvent, we optimized synthetic conditions, including reaction temperature and time, in the

coupling step to enhance the overall radiochemical yield and ratio of anomers (β/α). The newly developed method was applied for the radiosynthesis of 2'-deoxy-2'-[^{18}F]fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues to show the scope of the synthesis method. With the aim of preparing the Investigational New Drug (IND) application for [^{18}F]FMAU, three process validation batches of [^{18}F]FMAU were carried out using the new method. The resulting [^{18}F]FMAU tracer was then subjected to microPET imaging of tumor-bearing mice.

EXPERIMENTAL SECTION

Materials. 2-O-(Trifluoromethanesulfonyl)-1,3,5-tri-O-benzoyl- α -D-ribofuranose was either synthesized in accordance with the reported procedure³⁴ or obtained from ABX advanced biochemical compounds GmbH (Germany). [^{18}O]H₂O was purchased from Huayi Isotopes Co. All other chemicals and solvents were obtained from Sigma-Aldrich. 1,4-Dioxane (anhydrous, 99.8%) was tested for peroxide formation prior to use after opening the bottle. The ion exchange cartridges were obtained from ABX advanced biochemical compounds GmbH (Germany).

HPLC Methods. Analytical and semipreparative reversed-phase high-performance liquid chromatographies (HPLC) were carried out using two Thermo Scientific UltiMate 3000 HPLC systems. Semipreparative HPLC was performed using a Phenomenex Luna C18(2) reversed-phase column (5 μm , 250 mm $\times$ 10 mm). The flow rate was 3.5 mL/min with the isocratic mobile phase of 4% acetonitrile in water. The UV absorbance was recorded at 254 nm. Analytical HPLC was performed using a Phenomenex Luna C18(2) reversed-phase column (5 μm , 250 mm $\times$ 4.6 mm). The flow rate was 1 mL/min with the isocratic mobile phase of 8% acetonitrile in water with 0.1% trifluoroacetic acid (TFA). The UV absorbance was recorded at 254 nm. Model 101 and model 105 radiodetectors (Carroll & Ramsey Associates, Berkeley, CA) were used for the semipreparative and analytical HPLC system, respectively.

Radiosynthesis. [^{18}F]FMAU and its analogues were radiosynthesized in a semiautomatic synthesis module (Figure S1). The [^{18}F]fluoride ion was generated by the nuclear reaction [^{18}O] (p, n) [^{18}F] in a GE PETtrace 800 cyclotron. [^{18}F]fluoride ion in [^{18}O]water was transferred through a preconditioned QMA cartridge, and the retained [^{18}F]fluoride was eluted into a V-vial with a potassium carbonate solution (7.5 mg in 650 μL of deionized water). Kryptofix 222 solution (15.0 mg in 1.0 mL of anhydrous acetonitrile) was added to the V-vial, and the mixture solution was dried at 100 $^{\circ}\text{C}$ with nitrogen flow. Additional anhydrous acetonitrile was added to the V-vial and the reaction solution was azeotropically dried. The precursor 2-O-(trifluoromethanesulfonyl)-1,3,5-tri-O-benzoyl- α -D-ribofuranose solution (10.0 mg in 0.8 mL of anhydrous acetonitrile) was added to the dried [^{18}F] ion and heated at 85 $^{\circ}\text{C}$ for 20 min. Afterward, O,O'-bis-(trimethylsilyl)thymine (20 mg) or other 5-substituted uracil analogues, 200 μL of HMDS, 300 μL of 1,4-dioxane, and 150 μL of TMSOTf were added to the V-vial. The reaction solution was heated at 85 or 100 $^{\circ}\text{C}$ for various reaction times (15, 30, 45, and 60 min). After removing solvent, 400 μL of potassium methoxide solution (25% in methanol) and 400 μL of methanol were added. The mixture was heated at 85 $^{\circ}\text{C}$ for 5 min. After removing methanol, 6 N HCl was added to the reaction mixture. The crude reaction mixture was analyzed by analytical HPLC and purified by semipreparative HPLC. The

chemical purity and radiochemical purity of the final product were analyzed by HPLC. For the process validation batches of [^{18}F]FMAU, O,O'-bis(trimethylsilyl)thymine (20 mg), 200 μL of HMDS, 300 μL of 1,4-dioxane, and 150 μL of TMSOTf were used in the coupling step.

Quality Control for Process Validation Batches of [^{18}F]FMAU. All of the analytical test procedures were performed using high-quality solvents ($\geq 99.5\%$ purity), reagents, and materials which were carefully logged in, controlled, and verified in the same manner as the reagents for the manufacturing process. The drug product was assayed for total radioactivity using a qualified dose calibrator. The physical appearance of the drug product in the vial was determined by careful visual inspection under enough light. The final drug product in the vial must be clear and colorless without any visible particulates. Two samples totaling nominally ≥ 0.2 mL/sample were removed for quality control and the sterility test. The integrity of the sterilizing filter was tested. The filter was tested with increasing pressure applied by a calibrated gauge. The bubble point result must exceed the pressure of the manufacturer's specification to confirm filter integrity. The Kryptofix test was performed to demonstrate that the final product sample spot must show less intensity than the spot from the Kryptofix standard solution with a concentration of 50 $\mu\text{g/mL}$.

The retention time of standard FMAU was obtained using a certified standard produced by ABX advanced biochemical compounds GmbH (Germany). The radiochemical identity specification required the agreement of drug product and standard retention time within 0.5 min. The specification for the radiochemical purity was set up to be equal to or greater than 95%. The identity of [^{18}F]FMAU was validated by comparing the retention time of the nonradioactive FMAU standard and the [^{18}F]FMAU drug product. HPLC chromatography analysis was also applied to analyze chemical purity for the drug product. The specification of FMAU concentration was set up to be equal to or less than 8.33 $\mu\text{g/mL}$ based on our previous experience with [^{11}C]FMAU in nonhuman primates and humans. The amount of FMAU was calculated based on the FMAU UV peak area and the calibration curve. The total impurity in the [^{18}F]FMAU drug product was set up to be less than 3.6 $\mu\text{g/dose}$. This value includes only the unidentified impurities, that is, non-FMAU impurities.

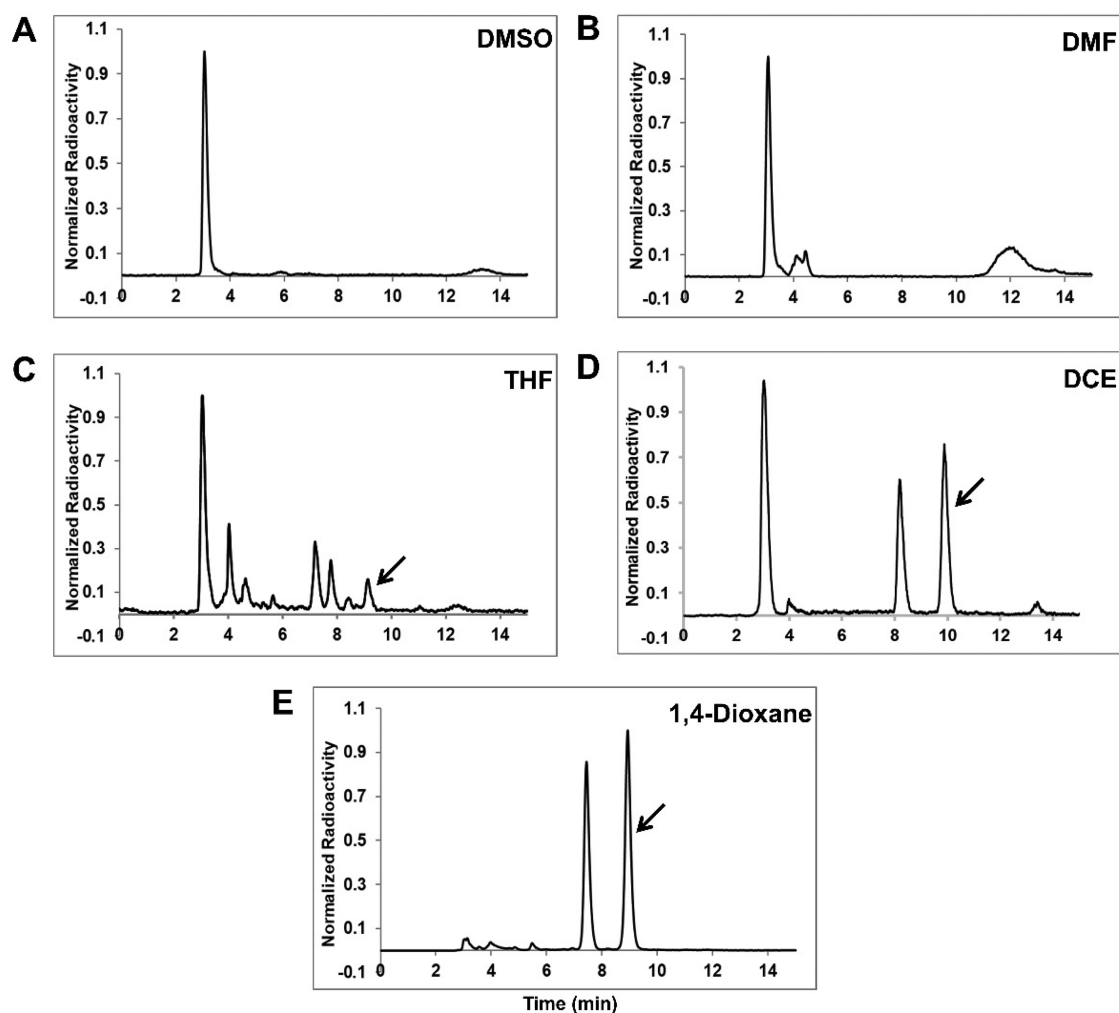
Residual solvent levels were determined using gas chromatography (GC). Methanol, acetonitrile, and 1,4-dioxane were used for the production of [^{18}F]FMAU and thus are potential residual solvent impurities. The permissible level of methanol, acetonitrile, and 1,4-dioxane in the final product must be equal to or less than 3000, 410, and 380 ppm, respectively as stated in the USP <467> residual solvent limits.

The radionuclidic identity of the final product was determined by measuring the half-life of the radionuclide in order to ensure it is [^{18}F]fluorine. This test was used to determine the identity of the radioactive nuclide of [^{18}F]fluorine in the sample of the final product. A sample was allowed to decay for a predetermined time and beginning and ending radioactivity measurements were compared and half-life calculated. The expected half-life of ^{18}F is 109.77 min. In the test to show radionuclidic identity, the half-life test result for ^{18}F must be between 105 and 115 min. The radionuclidic purity of the final product was determined by multichannel analysis (MCA). Photopeak energy for radioactive decay of [^{18}F]fluorine is 511 keV. Photopeak of the sample associated

Table 1. Solvent Effects on the Coupling of 2-Deoxy-2- ^{18}F fluoro-1,3,5-tri-*O*-benzoyl-D-arabinofuranose (^{18}F -Labeled Sugar) and *O,O'*-Bis(trimethylsilyl)thymine in the Radiosynthesis of ^{18}F FMAU (β -anomer)

solvent	polarity	class of residual solvents ^a	concentration limit (ppm) ^a	toxicity	% yield of β -anomer ^b	ratio of anomers (β/α)
DMSO	polar	3	5000	low	ND ^c	ND
DMF	polar	2	880	moderate	ND	ND
THF	polar	2	720	moderate	5.37	2.86
DCE	nonpolar	1	5	high	32.68	1.24
1,4-dioxane	nonpolar	2	380	moderate	48.07	1.06

^aData are cited from the United States Pharmacopeia (USP) General Chapter (467) Residual Solvents, Rev. 20190927. ^bRadiochemical yields (%). ^cND; not detected.

**Figure 2.** Analytical HPLC profiles of crude product in the radiosynthesis of ^{18}F FMAU using polar solvents: (A) dimethyl sulfoxide (DMSO), (B) *N,N*-dimethylformamide (DMF), (C) tetrahydrofuran (THF). Using nonpolar solvents: (D) 1,2-dichloroethane (DCE), (E) 1,4-dioxane. Arrows indicate the desired ^{18}F FMAU product (β -anomer).

with radioisotopic decay must be observed at the peak between 501 keV and 521 keV and possibly at 1.022 MeV (sum peak).

The specification of pH was set up to the range of 4.0–7.5. Bacterial endotoxin levels were tested using the Charles River Endosafe PTS system. The releasing specification for the bacterial endotoxin level is ≤ 17.5 EU/mL with a maximum injection volume of 6 mL. The 14-day sterility was tested using the direct inoculation method in which a sample was inoculated into two types of media within 30 h after synthesis of the drug product.

Partition Coefficient. The octanol–phosphate-buffered saline (PBS) partition coefficient was measured at room

temperature according to the previously reported procedure, and the value was designated as Log P.^{36,37} In brief, ^{18}F FMAU or other 5-substituted thymidine analogues (370 KBq) in 5 μL of PBS (pH = 7.4) was added to an Eppendorf tube including 500 μL of PBS (pH 7.4) and 500 μL of 1-octanol. The mixture was vortexed for 5 min and then centrifuged (12,500 rpm) for 8 min. The PBS and 1-octanol layers (200 μL of each layer) were pipetted into gamma-counter test tubes, respectively. The radioactivity was determined using a PerkinElmer 2480 WIZARD² automatic gamma counter (PerkinElmer Inc., Waltham, MA). The partition coefficients of 1-octanol-to-PBS were calculated as

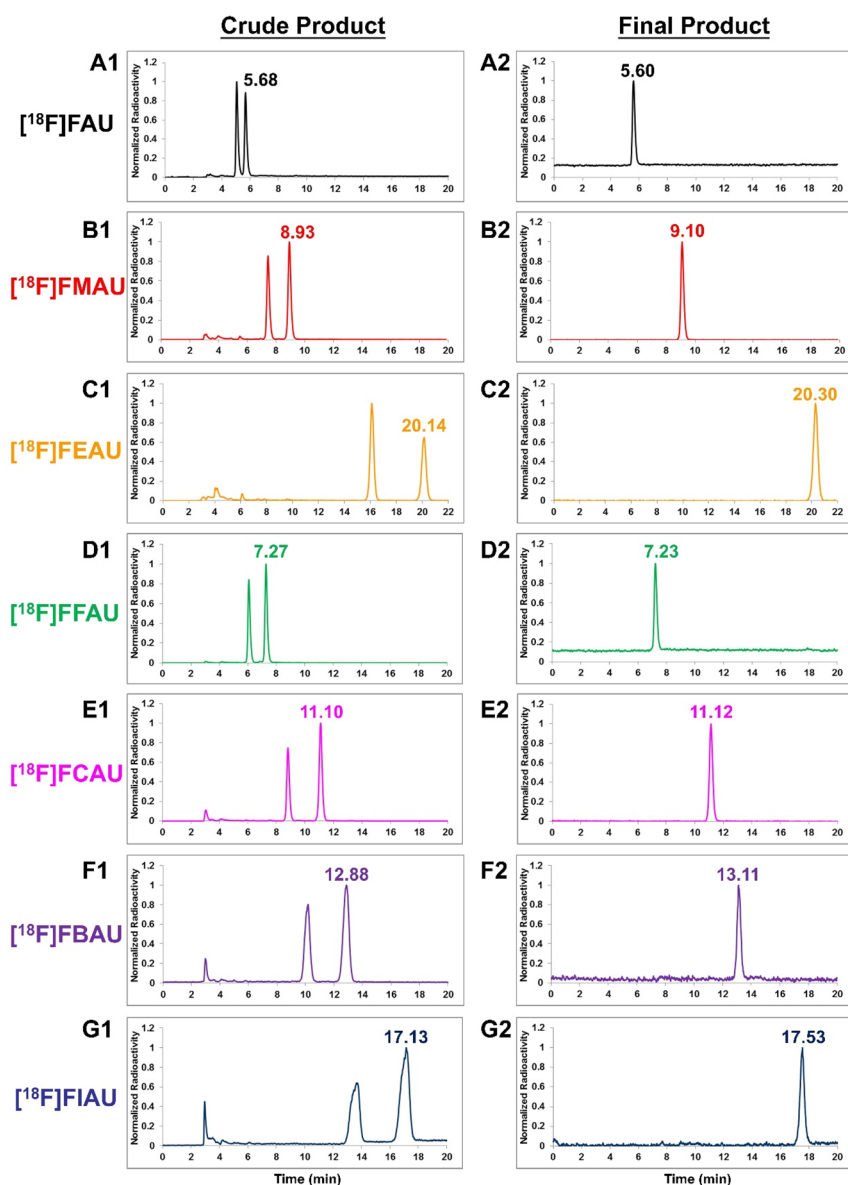


Figure 3. Analytical HPLC profiles of the crude product (A1–G1) and the final product (A2–G2) in the radiosynthesis of [^{18}F]FAU, [^{18}F]FMAU, [^{18}F]FEAU, [^{18}F]FFAU, [^{18}F]FCAU, [^{18}F]FBAU, and [^{18}F]FIAU using 1,4-dioxane as the solvent. The peaks labeled with retention time indicate the desired product (β -anomer).

$P = (\text{organic-phase cpm} - \text{background cpm}) / (\text{aqueous-phase cpm} - \text{background cpm})$, and the values were expressed as log P . Measurements were carried out in quintuplicate for each radiotracer.

Cell Culture. Both MDA-MB-231 human adenocarcinoma and U-87 MG human glioblastoma cell lines were purchased from American Type Culture Collection (Manassas, VA, USA). Tumor cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum at 37 °C in a humidified incubator containing 5% CO_2 .

Animal Tumor Models. All animal studies were approved by the Institutional Animal Care and Use Committee of the University of Southern California. Both MDA-MB-231 and U-87 MG tumor xenograft models ($n = 3/\text{group}$) were generated by subcutaneous injection of 5×10^6 tumor cells into the front right flank of female athymic nude mice (4–6 weeks old) purchased from Envigo Inc., Indianapolis, IN. The tumors

were permitted to grow 2–4 weeks until approximately 0.6–0.8 cm^3 in volume.

MicroPET Imaging. MicroPET scans were carried out using a rodent scanner (Siemens Inveon microPET scanner, Siemens Medical Solutions). About 7.4 MBq (200 μCi) of [^{18}F]FMAU was injected through the tail vein under isoflurane anesthesia condition. Five-minute static scans were obtained at 60 and 120 min postinjection (p.i.). The 3D-OSEM algorithm was applied for image reconstruction. For each microPET scan, the regions of interest (ROIs) were drawn over tumor, muscle, liver, and kidneys on the decay-corrected whole-body coronal images. The tumor-to-muscle (T/M), tumor-to-liver (T/L), and tumor-to-kidney (T/K) ratios were then calculated.

RESULTS AND DISCUSSION

Radiosynthesis of [^{18}F]FMAU and Its Analogues. Selection of appropriate solvents in PET drug manufacturing is of great importance for translating PET drugs into clinical

Table 2. Radiochemical Yield (%) and Analytical HPLC Retention Time of Crude and Final Product in the Radiosynthesis of [^{18}F]FAU, [^{18}F]FMAU, [^{18}F]FEAU, [^{18}F]FFAU, [^{18}F]FCAU, [^{18}F]FBAU, and [^{18}F]FIAU Using 1,4-Dioxane as the Solvent

radiotracer	% radiochemical yield ^a		ratio of anomers (β/α)	HPLC retention time (min)		
	α -anomer	β -anomer		crude product		final product
				α -anomer	β -anomer	β -anomer
[¹⁸ F]FAU	51.41	48.59	0.95	5.05	5.68	5.60
[¹⁸ F]FMAU	44.34	55.66	1.26	7.44	8.93	9.10
[¹⁸ F]FEAU	52.12	40.78	0.78	16.11	20.14	20.30
[¹⁸ F]FFAU	44.08	54.82	1.24	6.07	7.27	7.23
[¹⁸ F]FCAU	37.95	57.53	1.52	8.81	11.10	11.12
[¹⁸ F]FBAU	39.44	55.69	1.41	10.19	12.88	13.11
[¹⁸ F]FIAU	37.67	54.78	1.45	13.67	17.13	17.53

^aRadiochemical yield (%) is reported based on analytical HPLC.

use. In our previous effort of radiosynthesizing 2'-deoxy-2'-[^{18}F]fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues, we found that DCE can be used in the step of coupling ^{18}F -labeled sugar and 5-substituted uracil, in which the reaction was heated at 85 °C for 1 h to provide a β/α anomer ratio of 1.24:1 for the [^{18}F]FMAU synthesis (Table 1). However, DCE is listed as a Class 1 residual solvent in the USP, which is known to be highly toxic to humans, and thus is difficult to use in drug manufacturing for clinical investigations. With a goal of further improving the radiosynthesis of 2'-deoxy-2'-[^{18}F]fluoro-5-substituted-1- β -D-arabinofuranosyluracil

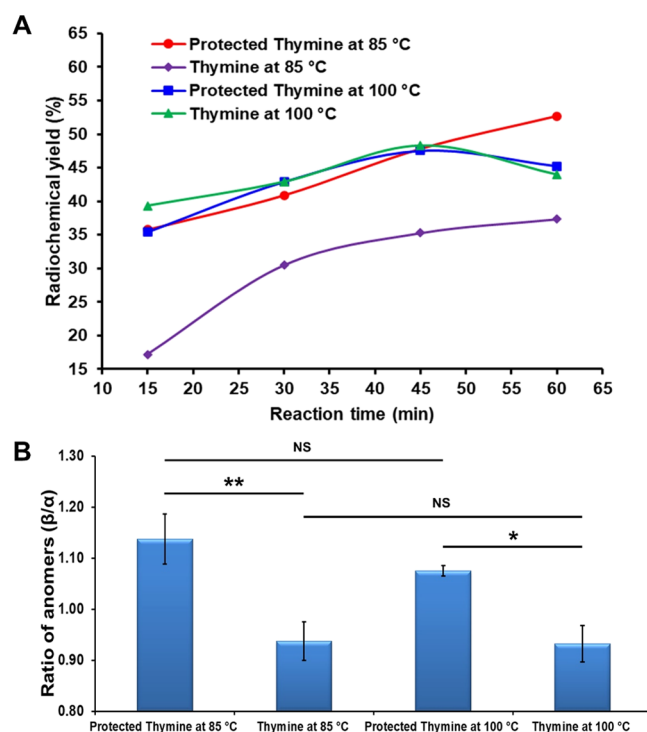


Figure 5. Coupling efficiency of 2-deoxy-2-[^{18}F]fluoro-1,3,5-tri-O-benzoyl-D-arabinofuranose (^{18}F -labeled sugar) and the protected thymine (*O,O'*-bis(trimethylsilyl)thymine) or thymine using 1,4-dioxane as the solvent at different reaction times and temperatures: (A) radiochemical yield (%) based on analytical HPLC; (B) ratio of anomers (β/α) at 60 min. Statistical significance between two groups is shown (* $P < 0.05$; ** $P < 0.01$; NS, nonsignificant).

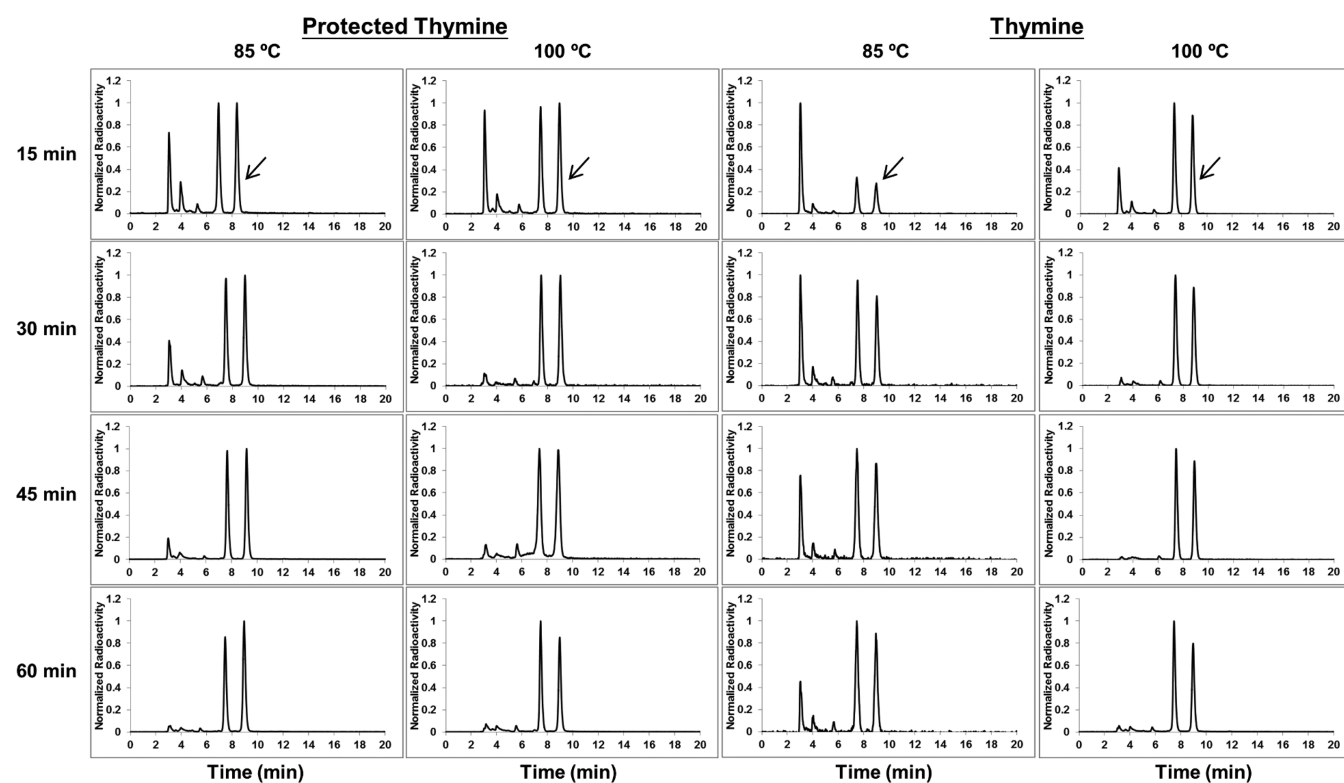


Figure 4. Analytical HPLC profile of crude [^{18}F]FMAU product using the protected thymine (*O,O'*-bis(trimethylsilyl)thymine) or thymine and 1,4-dioxane in the coupling step at different reaction times and temperatures. Arrows indicate the desired [^{18}F]FMAU product (β -anomer).

Table 3. Quality Control Data for Process Verification Batches of [¹⁸F]FMAU

QC test	release criteria	batch 1	batch 2	batch 3
radioactivity concentration at end of synthesis (mCi/mL)	1–75 mCi/mL	18.3838	8.3224	17.1949
final product appearance	clear, colorless, and free of particulates	pass	pass	pass
filter membrane integrity (bubble-point test) (psi)	≥50 psi	63	63	62
Kryptofix test	≤50 μg/mL	pass	pass	pass
radiochemical identity (HPLC)	within 0.5 min of the reference standard retention time	pass	pass	pass
		standard: 9.927 min	standard: 9.947 min	standard: 9.953 min
		sample: 10.002 min	sample: 10.083 min	sample: 10.102 min
radiochemical purity (HPLC) (%)	≥95%	99.40	100.00	99.44
chemical purity (FMAU mass, μg/mL)	≤8.33 μg/mL	1.6900	2.3075	4.3006
total impurity (non-FMAU impurities) ^a	<3.6 μg/dose	0.5775	1.5320	0.4868
residual solvents (GC) (ppm)	methanol: ≤ 3000 ppm acetonitrile: ≤ 410 ppm 1,4-dioxane: ≤ 380 ppm	Pass	Pass	Pass
radionuclidic identity (half-life)	between 105 and 115 min	109.5637	110.1338	109.3802
radionuclidic purity (KeV)	peak value is present between 501 and 521 keV	511.7	511.7	511.7
final product pH	4.0–7.5	5.0	5.0	5.0
bacterial endotoxin test	≤17.5 EU/mL with maximum dose volume 10 mL	<5 EU/mL	<5 EU/mL	<5 EU/mL
14-day sterility test	absence of microbials after 14-day incubation in two kinds of media	pass	pass	pass

^aTotal impurity value includes only the unidentified impurities, i.e. non-FMAU impurities.

Table 4. Measured 1-Octanol/PBS Partition Coefficients and log *P* Values of [¹⁸F]FAU, [¹⁸F]FMAU, [¹⁸F]FEAU, [¹⁸F]FFAU, [¹⁸F]FCAU, [¹⁸F]FBAU, and [¹⁸F]FIAU

radiotracer	partition coefficients of 1-octanol/PBS ^a	log <i>P</i>
[¹⁸ F]FAU	0.114 ± 0.011	−0.943 ± 0.041
[¹⁸ F]FMAU	0.265 ± 0.002	−0.577 ± 0.003
[¹⁸ F]FEAU	0.837 ± 0.035	−0.077 ± 0.018
[¹⁸ F]FFAU	0.113 ± 0.006	−0.952 ± 0.023
[¹⁸ F]FCAU	0.334 ± 0.022	−0.477 ± 0.030
[¹⁸ F]FBAU	0.430 ± 0.024	−0.367 ± 0.025
[¹⁸ F]FIAU	0.780 ± 0.024	−0.108 ± 0.013

^aMeasurements were carried out in quintuplicate for each tracer.

cil analogues and facilitating their clinical translation, we attempted to explore other solvents. Our investigation started with some polar solvents, such as dimethyl sulfoxide (DMSO) and *N,N*-dimethylformamide (DMF). Interestingly, using these polar solvents, we did not observe any desired products (Table 1 and Figure 2). After moving to nonpolar solvents, we found that the use of tetrahydrofuran (THF) can yield the desired products, but the radiochemical yield is minimal and unacceptable. Our continued efforts led to the identification of 1,4-dioxane, which is listed in Class II residual solvents with a residual concentration limit of 380 ppm. As compared to DCE with a 5 ppm concentration limit, 1,4-dioxane is considered a greener solvent. In addition, the employment of 1,4-dioxane as compared to DCE in the radiosynthesis of [¹⁸F]FMAU afforded an improved radiochemical yield (RCY) of the desired β -anomer product (48.07% vs 32.68%) (Table 1 and Figure 2). Encouraged by this result, we applied 1,4-dioxane for the radiosynthesis of other 2'-deoxy-2'-[¹⁸F]fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues, including [¹⁸F]FAU, [¹⁸F]FEAU, [¹⁸F]FFAU, [¹⁸F]FCAU, [¹⁸F]FBAU, and [¹⁸F]FIAU. Indeed, the results in Figure 3 and Table 2 show that our new method is quite versatile. The desired

product (β -anomer) can be clearly identified in the crude product as shown in analytical HPLC profiles (Figure 3). In addition, except for [¹⁸F]FCAU and [¹⁸F]FEAU, the RCY of the β -anomer is over 50% based on analytical HPLC, and the ratio of β/α anomers is greater than 1 in the radiosynthesis of [¹⁸F]FMAU, [¹⁸F]FFAU, [¹⁸F]FCAU, [¹⁸F]FBAU, and [¹⁸F]FIAU (Table 2). It is worth noting that the RCY of the β -anomer in the current study using 1,4-dioxane is significantly higher than what was reported previously using DCE,³⁸ suggesting that 1,4-dioxane as a coupling solvent is more effective in the radiosynthesis 2'-deoxy-2'-[¹⁸F]fluoro-5-substituted-1- β -D-arabinofuranosyluracil analogues.

To further optimize the conditions and improve the coupling efficiency and radiochemical yield, we utilized [¹⁸F]FMAU as an example to investigate the coupling step in the presence of 1,4-dioxane by changing various reaction factors, including reaction time (15, 30, 45, and 60 min), reaction temperature (85 and 100 °C), and the protected thymine versus thymine. The results are shown in Figures 4 and 5. As a function of reaction time, the coupling efficiency is increased overall. For instance, in the case of the protected thymine and reaction temperature at 85 °C, the RCY of the β -anomer was enhanced from 35.77% to 52.66% (Figure 5A). Interestingly, the RCY of the β -anomer for both the protected thymine and thymine at 100 °C is decreased after heating the reaction 15 min longer (from 45 to 60 min), indicating that appropriate reaction time is important for the coupling step at 100 °C. In addition, fixing the reaction time at 45 min, we did not observe significant changes of the RCY of the β -anomer at 100 °C for the protected thymine and thymine, suggesting that protection of thymine may not be critical for the RCY at 100 °C. However, a significant RCY improvement was observed at 85 °C for 45 min using the protected thymine versus thymine (47.79% vs 35.29%). Similarly, an enhanced RCY was yielded at 85 °C for 60 min using the protected thymine versus thymine (52.66% vs 37.36%). Furthermore, we also calculated

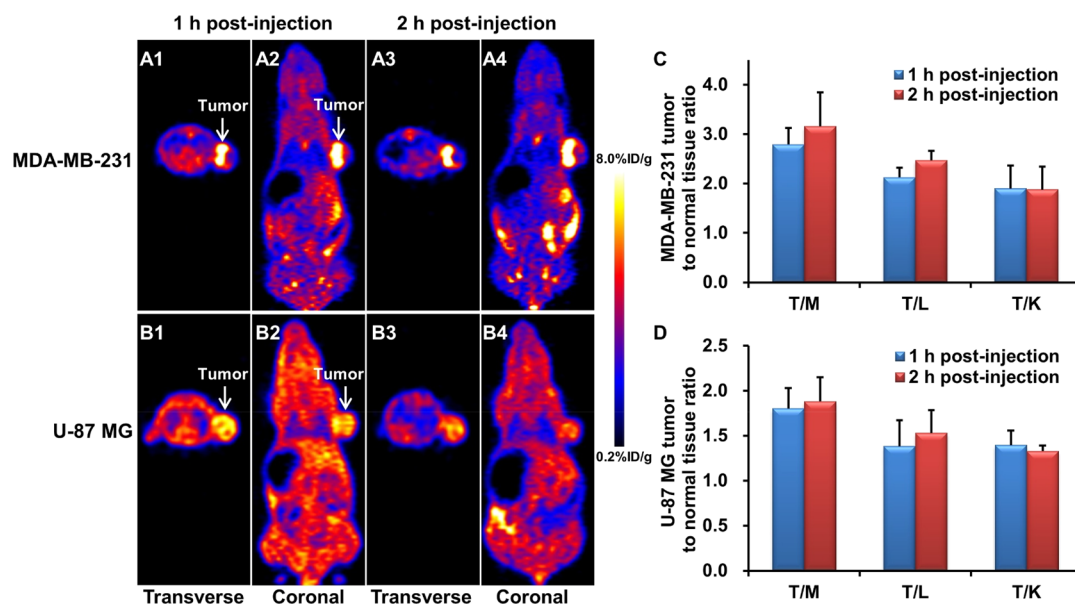


Figure 6. Representative microPET images of subcutaneous MDA-MB-231 (A1–A4) and U-87 MG (B1–B4) tumor-bearing nude mice at 1 and 2 h postinjection (p.i.) of $[^{18}\text{F}]\text{FMAU}$. Tumor-to-muscle (T/M) ratio, tumor-to-liver (T/L) ratio, and tumor-to-kidney (T/K) ratio of $[^{18}\text{F}]\text{FMAU}$ at 1 and 2 h p.i. with mouse xenograft models bearing subcutaneous MDA-MB-231 (C) or U-87 MG (D) tumor. Arrows indicate tumors.

the ratio of β/α anomers based on the analysis of analytical HPLC for the coupling reaction at 60 min. As shown in Figure 5B, the ratio of β/α anomers for the protected thymine is significantly higher than that of thymine at 85 °C (1.14 ± 0.05 vs 0.94 ± 0.04) and 100 °C (1.08 ± 0.01 vs 0.93 ± 0.04), demonstrating that using the protected thymine is critical to obtaining a higher ratio of β/α anomers. Nonsignificant changes in the ratio of β/α anomers were observed for both the protected thymine and thymine at different temperatures (85 °C vs 100 °C). Taken together, based on the results of the β -anomer RCY, the ratio of β/α anomers, and the length of reaction time, we determined that using the protected thymine and heating at 85 °C for 60 min is the best condition for the coupling step in the radiosynthesis of $[^{18}\text{F}]\text{FMAU}$. The semipreparative HPLC UV and radioactivity profiles of crude $[^{18}\text{F}]\text{FMAU}$ product using the newly developed method are presented in Figure S2. The analytical HPLC UV and radioactivity for coinjection of cold authentic anomers (α - and β -anomer) and $[^{18}\text{F}]\text{FMAU}$ are displayed in Figure S3.

Quality Control for Process Validation Batches of $[^{18}\text{F}]\text{FMAU}$. Using the optimized condition in the presence of 1,4-dioxane, we performed three consecutive process validation batches of $[^{18}\text{F}]\text{FMAU}$ to fulfill the requirements of the IND application. Quality control testing of the $[^{18}\text{F}]\text{FMAU}$ product was conducted according to the guidelines outlined in the USP and as described previously in the method section. Testing included visual inspection, pH, residual Kryptofix 222, chemical purity and radiochemical purity, specific activity, radionuclidic identity and purity, sterile filter integrity, bacterial endotoxin analysis, and sterility testing. Results for three process verification batches are reported in Table 3. All validation batches for process verification passed all required criteria for release. The results based on the new method of using 1,4-dioxane for $[^{18}\text{F}]\text{FMAU}$ manufacture satisfy the requirements of the submission of the IND application.

Partition Coefficient. To evaluate the hydrophilicity of PET tracers, we measured the 1-octanol/PBS partition coefficient value as expressed as log P . The log P values of

$[^{18}\text{F}]\text{FAU}$, $[^{18}\text{F}]\text{FMAU}$, $[^{18}\text{F}]\text{FEAU}$, $[^{18}\text{F}]\text{FFAU}$, $[^{18}\text{F}]\text{FCAU}$, $[^{18}\text{F}]\text{FBAU}$, and $[^{18}\text{F}]\text{FIAU}$ were determined to be -0.943 ± 0.041 , -0.577 ± 0.003 , -0.077 ± 0.018 , -0.952 ± 0.023 , -0.477 ± 0.030 , -0.367 ± 0.025 , and -0.108 ± 0.013 , respectively (Table 4). The log P values suggest that the hydrophilicity is gradually reduced when the 5-hydrogen of 2'-deoxy-2'- $[^{18}\text{F}]\text{fluoro-1-}\beta$ -D-arabinofuranosyluracil is substituted by fluoro, methyl, chloro, bromo, iodo, and ethyl groups, respectively. The hydrophilicity of these analogues determined by log P showed a similar pattern in general as appeared at the retention times on the analytical HPLC (Figure 3 and Table 2).

PET Imaging. We next studied the tumor PET imaging of $[^{18}\text{F}]\text{FMAU}$ in animals. For this purpose, we selected two aggressive tumor cell lines, MDA-MB-231 triple-negative breast cancer cell line and U-87 MG glioblastoma cell line, to establish tumor xenografts in mice. After the intravenous injection of $[^{18}\text{F}]\text{FMAU}$ at 1 and 2 h, the mice ($n = 3/\text{group}$) were scanned through a microPET imaging system. The representative decay-corrected transverse and coronal sections that contained the tumors at 1 and 2 h postinjection (p.i.) are displayed in Figure 6 A1–A4 (MDA-MB-231 tumor model) and B1–B4 (U-87 MG tumor model). For microPET scans, radioactivity accumulations in tumors and major tissues/organs were quantified by calculating the ROIs that comprised the entire organ on the coronal images. For the MDA-MB-231 tumor model, tumor uptake of $[^{18}\text{F}]\text{FMAU}$ was calculated to be 6.4 ± 0.4 and $7.2 \pm 0.6\%$ ID/g at 1 and 2 h p.i., respectively. The ratio of MDA-MB-231 tumor uptake to muscle, liver, and kidney uptake was calculated to be 2.8 ± 0.3 , 2.1 ± 0.2 , and 1.9 ± 0.5 (at 1 h p.i.), and 3.2 ± 0.7 , 2.5 ± 0.2 , and 1.9 ± 0.5 (at 2 h p.i.), respectively. For the U-87 MG tumor model, tumor uptake of $[^{18}\text{F}]\text{FMAU}$ was calculated to be 6.0 ± 0.2 and $5.6 \pm 0.4\%$ ID/g at 1 and 2 h p.i., respectively. The ratio of U-87 MG tumor uptake to muscle, liver, and kidney uptake was calculated to be 1.8 ± 0.2 , 1.4 ± 0.3 , and 1.4 ± 0.2 (at 1 h p.i.), and 1.9 ± 0.3 , 1.5 ± 0.3 , and 1.3 ± 0.1 (at 2 h p.i.), respectively. At 1 h versus 2 h p.i., nonsignificant changes were

observed for the ratio of T/M, T/L, and T/K in both tumor models. At all imaging time points, tumors were clearly visible with good contrast to the background. We noticed that the tumor uptake values of [^{18}F]FMAU are different between the MDA-MB-231 and U-87 MG tumor models. This is presumably due to the different stages of tumor cell proliferation, which we observed in animal studies of [^{18}F]FMAU with prostate tumors.¹⁵ Overall, we believe that the newly developed radiosynthesis method of [^{18}F]FMAU and its analogues will facilitate future investigations in both preclinical and clinical studies.

CONCLUSIONS

We have successfully synthesized 5-substituted 2'-deoxy-2'-[^{18}F]fluoro-arabino-furanosyluracil analogues in excellent radiochemical purity using an improved synthesis method. 1,4-Dioxane is a less-toxic alternative to DCE that also provides better radiochemical yields. The use of a less toxic solvent allows for the translation of the improved approach to clinical production. This new method is versatile, which permits a broad range of use for ^{18}F -labeling of other nucleoside analogues.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acspsci.0c00184>.

Schematic of radiosynthesis module, semipreparative HPLC profiles, and analytical HPLC profiles (PDF)

AUTHOR INFORMATION

Corresponding Author

Kai Chen – Department of Radiology, Keck School of Medicine, University of Southern California, Los Angeles, California 90033, United States; orcid.org/0000-0002-8647-1182; Email: chenkai@med.usc.edu

Authors

Jindian Li – Department of Radiology, Keck School of Medicine, University of Southern California, Los Angeles, California 90033, United States

Juno Van Valkenburgh – Department of Radiology, Keck School of Medicine, University of Southern California, Los Angeles, California 90033, United States

Peter S. Conti – Department of Radiology, Keck School of Medicine, University of Southern California, Los Angeles, California 90033, United States

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acspsci.0c00184>

Notes

The authors declare no competing financial interest.

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